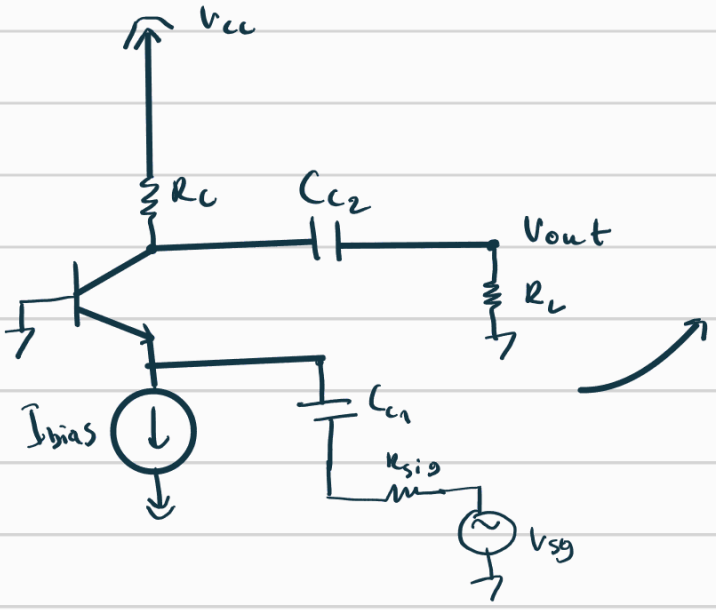
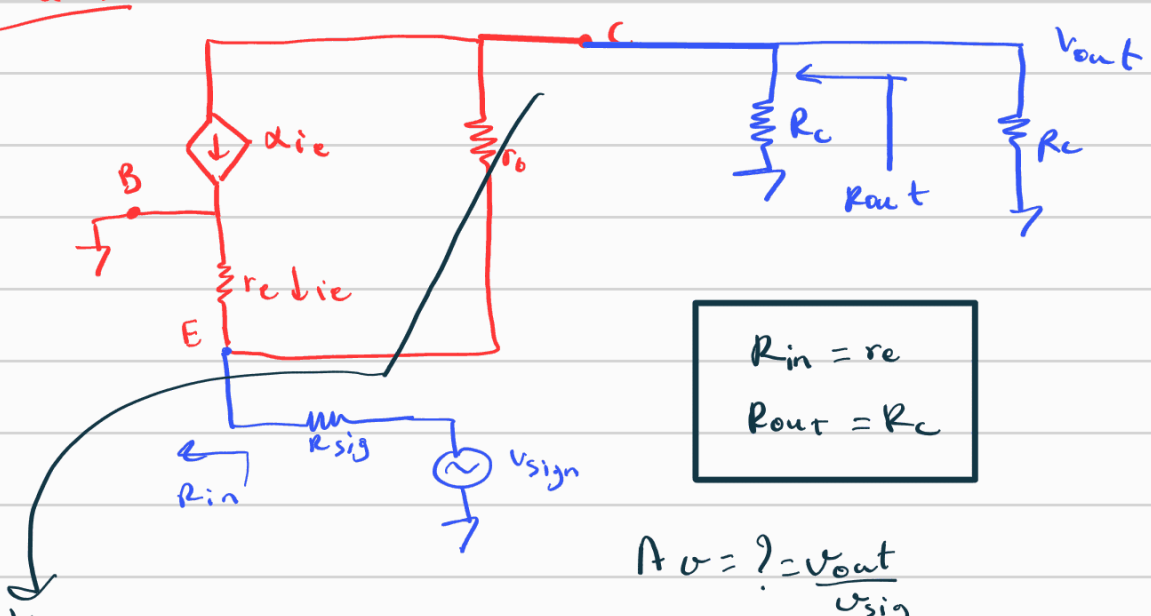
Common-Base Amplifier



T model



$R_{in} = r_e$

$R_{out} = R_C$

$A_v = ? = \frac{v_{out}}{v_{sig}}$

$V_A = \infty$

$r_o = \infty$

$$\frac{v_{out} = -\alpha \cdot i_e (R_C // R_L)}{v_{sig} = i_e (R_{sig} + r_e)}$$

$$= \frac{\alpha (R_C // R_L)}{R_{sig} + r_e}$$

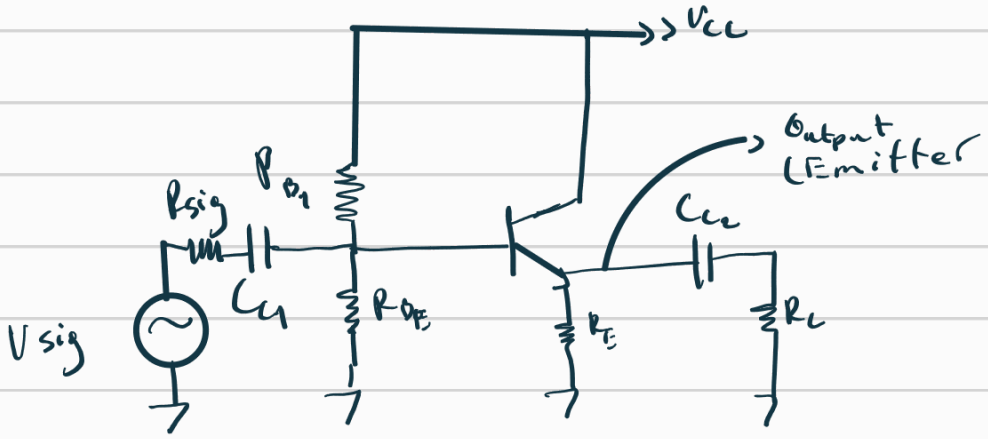
$\Rightarrow$ if $R_{sig} \ll r_e$ ignore R_{sign}

$\Rightarrow g_m (R_C // R_L)$ $\left[r_e = \frac{1}{g_m} \right]$

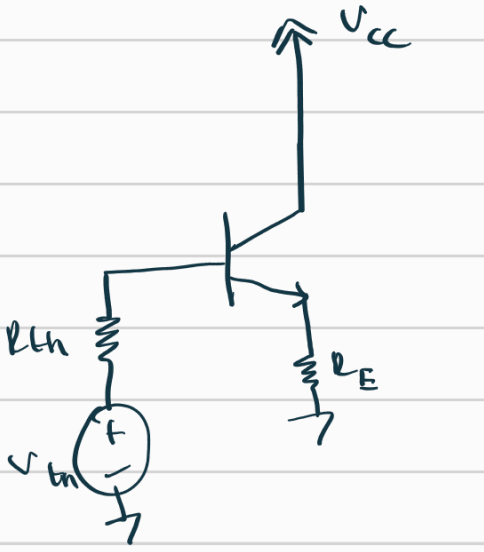
- Small input Resistance → normalde yüksek giriş dirençisteriz sinyalin zayıflamaması için (opposite C-E)
- Moderate - High Output Resistance (C-E)
- High gain (C-E)

(Emitter- Follower)

Common - Collector Amplifier

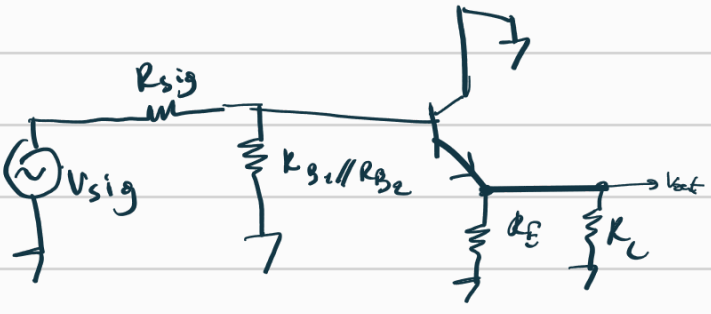


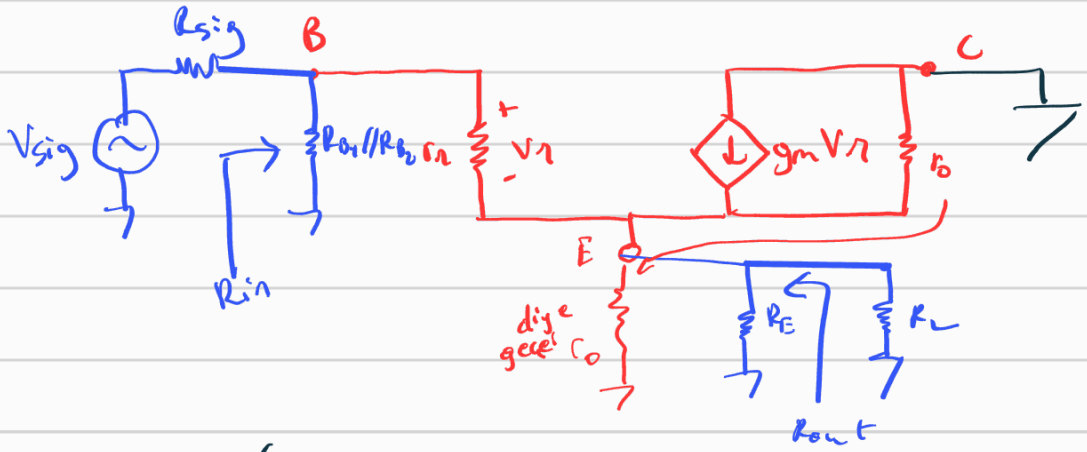
DC



AC

A ≠ ∞





$R_{in} = ?$ $\quad (R_{B1} // R_{B2}) // (r_\pi + (\beta+1)(R_E // R_L // r_o))$

$R_{out} = ? \; r_o // R_E // \left(\dfrac{(R_{sig} // R_{B1} // R_{B2}) + r_\pi}{\beta + 1} \right)$

$\dfrac{V_{out}}{V_{sig}} = \dfrac{V_\pi (g_m + 1/r_\pi) R_L'}{V_\pi + V_\pi (g + 1/r_\pi) R_L'}$ $\quad R_{sig} = 0$ (Assumption) $\Rightarrow$

$V_{sig} = V_\pi + V_\pi (g + 1/r_\pi) R_L'$

$$A_v = \frac{V_{out}}{V_{sig}} = \frac{(g_m + 1/r_\pi) R_L'}{1 + (g_m + 1/r_\pi) R_L'}$$

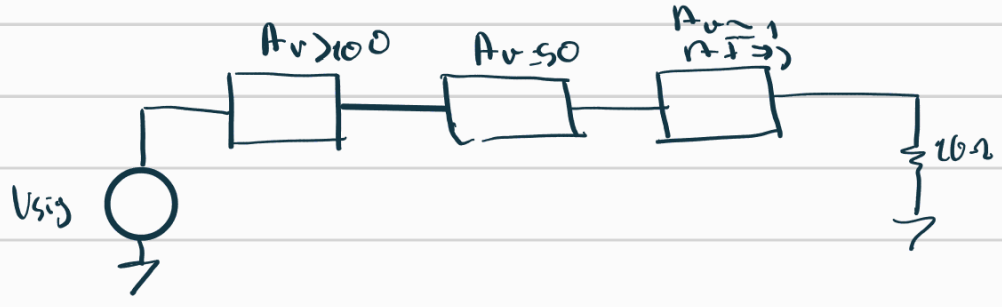
Her zaman

- Voltage gain < 1 $\quad (0,6 \sim 0,9)$
- Highcurrent output gain
- Input High Resistance
- Low Output resistance

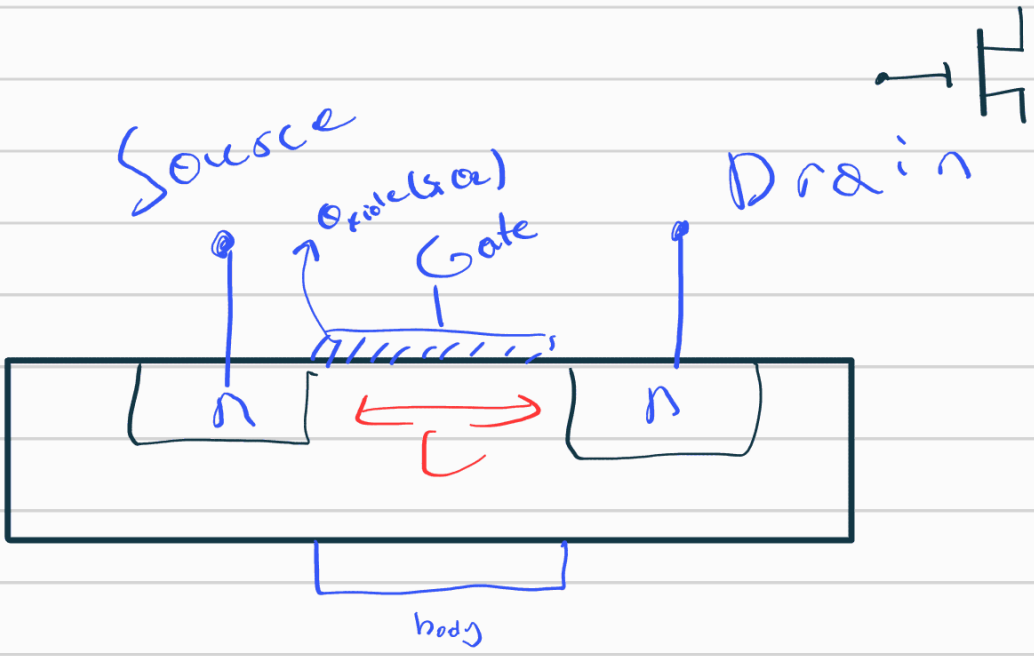
You can use as

Voltage buffer

- Use it on the last stage of ripple stage amplifier

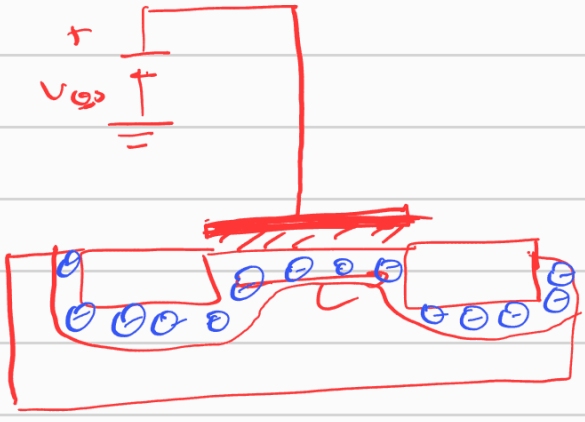


Mosfet



Drain is positive

5.1.3 Creating a channel for current Flow



Treshold Voltage (V_T) - is the min value of v_{gs}

field effect!

effective/ overdrive voltage

$$v_{ov} = v_{GS} - V_t$$

oxide capacitance (C_{ox})

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}} \text{ in } F/m^2 \qquad C = \frac{\varepsilon A}{d}$$

$$|Q| = C_{ox}(WL)\, v_{ov} \text{ in } C$$

Applying a small v_{ds}

$$i_D = (C_{ox} W v_{ov}) \left(\frac{\mu_n v_{DS}}{L}\right) A$$

? kontrol ex

⋮

small v_{DS}

(eq 5.7) $$i_D = \left[(\mu_n C_{ox}) \frac{W}{L} v_{ov}\right] v_{DS} \text{ in } A$$

(eq 5.8) $$r_{DS} = \frac{v_{DS}}{i_D} = \frac{1}{(\mu_n C_{ox})\left(\frac{W}{L}\right)} \text{ ---}$$

Operation as v_{DS} increased

$$v_{GD} = v_{GS} - v_{DS}$$

→ v



Saturation Current formula

$$I_D = \underbrace{\frac{1}{2}\mu_n C_{ox}\frac{W}{L}}_{k_n'} (V_{GS}-V_t)^2$$

$k_n = \mu_n C_{ox}\frac{W}{L}$

$$\boxed{I_D = \mu_n C_{ox}\frac{W}{L}\left((V_{GS}-V_t)-\frac{1}{2}V_{DS}\right)V_{DS}}$$

Cutoff | Triode | saturation

$V_{GS} < V_T$

$V_{GS} \geq V_T$, $V_{DS} < V_{OV}$

$V_{GS} \geq V_T$, $V_{DS} \geq V_{OV}$

Operation for $V_{DS} \geq V_{OV}$

The p Channel MOSFET

$V_{SG} \geq |V_{tp}|$ → $V_{tp} < 0$

$V_{SG} < |V_T|$ → cutoff

$V_{SG} \geq |V_T|$ $\Rightarrow$ Saturation

$V_{SD} \geq V_{OV}$ $\Rightarrow$

$V_{SG} \geq |V_t|$
$V_{SD} < V_{OV}$ } Triode

(NMOS) $I_{D1} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{OV})^2$

(PMOS) $I_{D2} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{OV})^2$

if $\frac{W}{L}$, C_{ox}, V_{OV} are equal $\Rightarrow I_{D1} > I_{D2}$ because

$\mu_n > \mu_p$

↓ electron mobility

Quick Recap

The $i_D - V_{GS}$ Characteristic

In sat

Ex 5.3

$I_D = 0,1 mA$

$V_D = 0,5 V$

$N_n C_{ox} = 100 \mu A/V^2$ $L = 1 \mu m$, $W = 32 \mu m$,

$0,4 mA = \frac{1}{2} \; 0,1 mA \left(\frac{32}{1}\right) (V_{OV})^2$

$V_{OV} = 0,5 \Leftarrow (V_{OV})^2 = 0,25$ $\rightarrow V_{OV} = +0,5$

$V_{GS} = 0.5 + 0.7 \rightarrow V_T$ $\quad V_S = -1.2$

$R_S =$

⋮

Ex 5.5

$V_{GS} \geq V_T$

$5 - 0 \geq 1V$

$V_{DS} \geq V_{OV}$

$0.1 \geq 4V$? ✗ Triode

$I_D = \mu_n C_{ox} \frac{W}{L} \left(V_{OV} - \frac{1}{2} V_{DS}\right) V_{DS}$

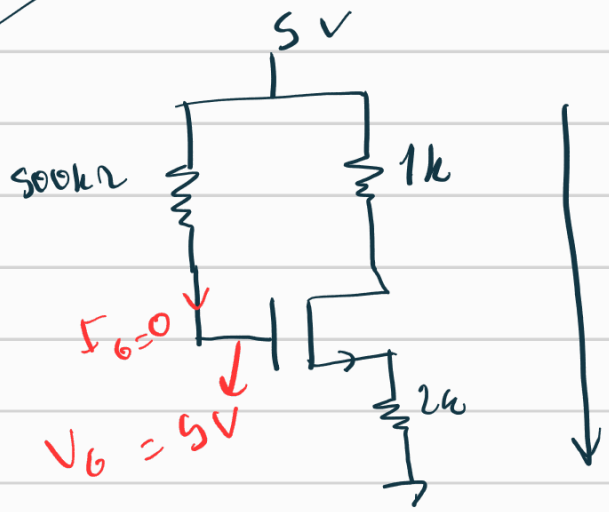
$r_{DS} = \frac{V_{DS}}{I_D}$

$1 mA/V^2 \left(4 - 0.05\right) 0.1$

$I_D = 0.395\ mA$

$\frac{V_{DS}}{I_D} = 253\Omega$

Örnek



$\mu_n C_{ox} \frac{W}{L} = 0.25 mA/V^2$

$V_{tn} = 2V$

$\lambda = 0 \quad V_A = 0V$

$I_D, V_G, V_S, V_D = ?$

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)^2$$

$$I_D = \frac{1}{2} 0{,}25 \frac{mA}{V^2} (5 - V_S - 2)^2$$

$$I_D = \frac{1}{2} 0{,}25 \frac{mA}{V^2} (3 - V_S)^2$$

$$I_D = \frac{1}{2} 0{,}25 \frac{mA}{V^2} (3 - 2I_D)^2$$

$$I_D = \frac{1}{2} \frac{0{,}25\,mA}{V^2} (9 + 4I_D^2 - 6I_D)$$

$$8 I_D = 9 + 4I_D^2 - 6I_D$$

$$4I_D^2 - 14 I_D + 9 = 0$$